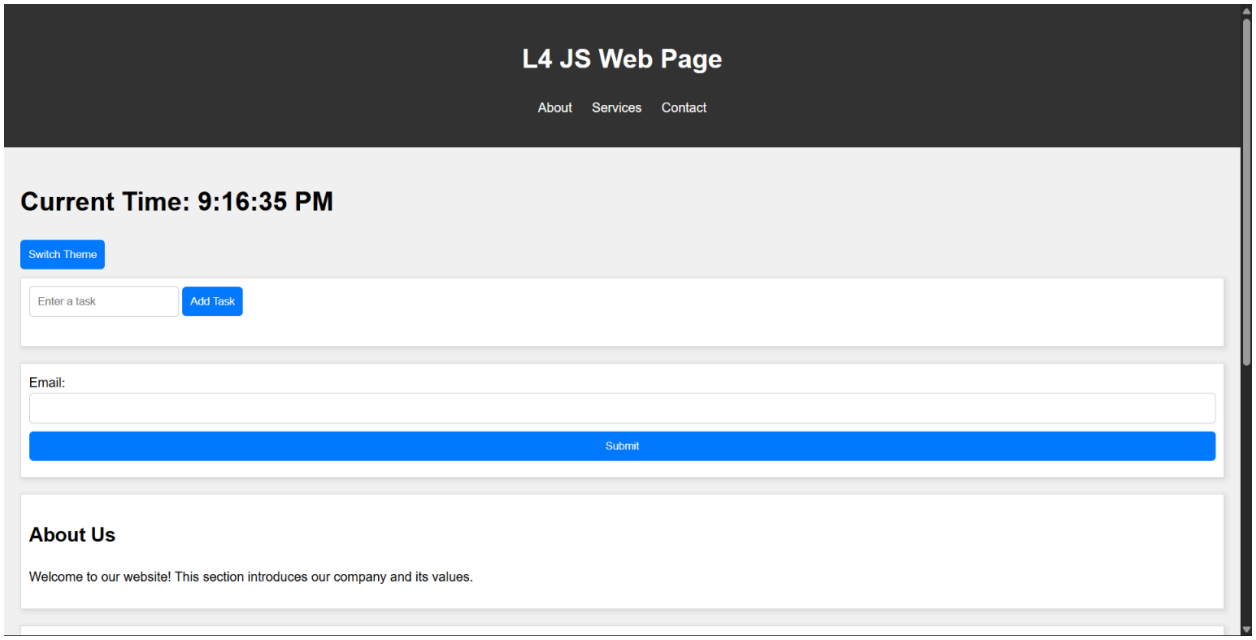



Integrating JavaScript functionality shifted the focus from static design to creating dynamic, user-driven web interactions. Implementing event listeners and managing the Document Object Model provided practical experience in updating elements dynamically based on user actions. Handling state changes, such as theme toggling and local storage persistence, initially required careful debugging of script execution timing. Resolving these logic hurdles highlighted the importance of clean event handling and proper state management in client-side development. Finally, combining interactive features with the existing design elements successfully enhanced the overall usability and responsiveness of the portfolio interface.